

Programming Notes - Functions and Input Validation

Course: Computer Science

Why Functions?

Functions divide a program into reusable pieces. They reduce repetition, improve readability, and make testing easier. A function may accept parameters and may return a value.

Example

```
def calculate_tax(price, rate):  
    return price * rate
```

Calling `calculate_tax(100, 0.13)` returns 13.0.

Input Validation

Never assume user input is valid. Programs should check values before using them. Validation can reject impossible values, ask again, or provide a clear error.

Example Strategy

If an age must be between 0 and 120, convert the input carefully, verify the range, and only continue when the value is valid.

Design Principle

Keep each function focused on one clear responsibility. A function called `calculate_total` should calculate the total; it should not also display an entire menu and save files.

Practice

- Write a function `area_rectangle(length, width)` that returns the area.
- Explain the difference between a parameter and an argument.
- Describe how you would validate a percentage that must be from 0 to 100.